Innovation Center for Education

THE YENEPOYA INSTITUTE OF ARTS SCIENCE COMMERCE AND MANAGEMENT

PROJECT-LOW LEVEL DESIGN ON

Cloud Forensics Automation for Retail Breaches

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TABLE OF CONTENT :

1. Introduction ---------- 2
1.1 Purpose and
Scope ----------2
1.2 Project Executive
Summary ----------3
1.3 Constraints ----------3
1.4 Document Organization ----------3
2. System Overview ----------4
3. Module Design ----------4
3.1 File Integrity Scanning Module ----------4
3.2 Baseline Management Module ----------4
3.3 File Integrity Analysis Module ----------4
3.4 Retail Data Management Module ----------4
3.5 Retail Simulation Module ----------5
3.6 Comparison & Detection Module ----------5
3.7 Alert & Risk Detection Module ----------5
3.8 Reporting Module ----------5
3.9 Visualization & Dashboard Module ----------6
3.10 Automation Control Module ----------6
4. Data Flow Design ----------6
5. Database Design ----------6
6. User Interface Design ----------7
7.Security Implementation ----------7
8. Testing and Validation ----------7
9. System Demonstration / Working ----------8
9. System Demonstration / Working ----------9-11

1. INTRODUCTION

1.1 Purpose and Scope

Purpose:

This document provides a detailed low-level design of a Cloud Forensics Automation system for Retail Breaches. The system is designed to automate the collection of file metadata such as creation and last modified timestamps to detect possible file tampering during a retail data breach.
The system simulates forensic investigation in cloud or retail storage environments by analyzing file history.

Scope:

The system covers:

- Automated file system scanning
- Metadata extraction (creation & modification time)
- Logging forensic evidence into a CSV report
- Simulation of cloud forensic investigation

In Scope:

- Local/cloud directory scanning(simulated)
- File metadata collection
- CSV report generation
- Evidence logging for investigation

Out of Scope

- Real-time cloud monitoring
- Integration with AWS/Azure APIs
- Deep file content analysis
- Live incident response systems

1.2 Project Executive Summary

Retail cloud environments store critical business and customer data. During a cyberattack, attackers may modify or delete files to hide traces. Digital forensics plays a key role in identifyingsuch changes.
This project introduces a system that:

- Scans directories automatically
- Extracts file metadata (creation & last modified time)
- Logs all file details into a structured CSV report
- Simulates cloud forensic evidence collection

Technologies Used:
- Python
- OS module
- Datetime module
- CSV module

1.3 Constraints

Technical Constraints

- Depends on OS-level timestamps
- Limited to metadata analysis
- Performance varies with file volume

Security Constraints

- Must not alter any file
- Logs must remain accurate and secure

Operational Constraints

- Designed for academic/demo use
- Cloud environment is simulated locally

1.4 Document Organization

The document is structured into:

1. Introduction
2. System Overview
3. Module Design
4. Data Flow
5. Data Structure
6. Interface Design
7. Security Implementation
8. Testing Strategy

2. System Overview

The system is a web-based forensic application designed to detect file tampering in a retail environment. It provides an interactive dashboard where users can scan files, create baselines, simulate retail operations, and analyze file integrity.
The system compares current file metadata with previously stored baseline data to identify suspicious changes such as modifications, additions, or deletions.
The system also generates alerts and forensic reports based on detected changes, helping investigators quickly identify potential security breaches

3. Module Design

3.1 File Integrity Scanning Module

- Scans all files in the retail directory
- Extracts metadata:
  - File name
  - Size
  - Created time
  - Modified time
- Displays scan results in table format
- Updates system status

3.2 Baseline Management Module

- Stores initial file state as baseline
- Loads baseline for comparison
- Used to detect future changes
- Ensures accurate forensic comparison

3.3 File Integrity Analysis Module

- Compares current scan with baseline
- Classifies files as:
  - Normal
  - Modified
  - New
  - Deleted
- Calculates risk level (Low/Medium)
- Displays results in structured table

3.4 Retail Data Management Module

- Allows:
  - Add customer
  - Edit customer
  - Delete customer
- Simulates real retail database
- Changes reflect in file system (like customers.csv)

3.5 Retail Simulation Module

- Simulates real-world retail operations
- Features:
  - Add customer
  - Update product price
- Triggers file changes
- Used to test tampering detection

3.6 Comparison & Detection Module

- Compares baseline with current data
- Generates:
  - Comparison summary
  - Change distribution graph
- Identifies:
  - Modified files
  - Unchanged files
- Displays visual analytics

3.7 Alert & Risk Detection Module

- Detects suspicious file changes
- Generates alerts:
  - “1 Suspicious File Detected”
- Assigns risk levels:
  - Low
  - Medium
- Displays alert details

3.8 Reporting Module

- Generates forensic reports in CSV
- Includes:
  - File changes
  - Timestamps
  - Risk levels
- Suppor ts:
  - Evidence report download
  - Comparison report download

3.9 Visualization & Dashboard Module

- Displays:
  - Total files
  - Normal files
  - High-risk files
  - File size
- Shows:
  - Bar charts
  - Timeline charts
- Helps in quick analysis

3.10 Automation Control Module

- Controls full workflow:
  - Scan → Baseline → Compare → Alert → Report
- Ensures smooth system execution
- Connects all modules

4. DATA FLOW

Write this exactly:

i. User clicks “Scan Retail Files”
ii. System scans directory and extracts metadata
iii. User sets baseline data
iv. User performs retail operations (add/edit data)
v. Files are modified in the system
vi. User runs scan again
vii. System compares current data with baseline
viii. Changes are detected and classified
ix. Alerts are generated for suspicious files
x. Evidence report is created and available for download

5. DATA STRUCTURE

Now you clearly have more fields:

- file_name
- file_path
- size
- created_time
- modified_time
- status
- risk_level

6. User Interface Design

The system provides a web-based dashboard interface for users to interact with the forensic system. It includes features such as file scanning, baseline management, data simulation, and result visualization.
The interface includes:

- File Integrity Scanning section
- File Integrity Analysis table
- Retail Data Panel (Customer management)
- Retail Simulation Panel
- Summary statistics and charts
- Alert and reporting sections

The interface is designed to be simple, interactive, and user-friendly.

7. Security Implementation

The system ensures basic security for forensic integrity:

- Read-only scanning of files (no direct modification)
- Baseline data stored securely for comparison
- Detection of unauthorized file changes
- Alerts generated for suspicious activity
- Evidence reports generated without altering original data

8. Testing and Validation

The system was tested using multiple scenarios to verify its functionality:

- File scanning tested on retail dataset
- Baseline creation and loading validated
- File modification detection verified
- Customer data changes tested for tampering detection
- Alert system tested for suspicious file detection
- CSV report generation validate
- The system successfully detected file changes and generated accurate forensic reports.

9. Demonstration

Step 1 : **InitialScan**
System scansretail files → shows:
- Total Files:3
- All files normal

Step 2 : **Baseline Creation**
User sets baseline → system stores original state

Step 3 **: Simulation**
User adds new customer (e.g., “robert”)
customers.csv gets modified

Step 4 **: Re-scan**
System detects:
- 1 Modified file
- Risk Level: Medium

Step 5 : **Alert Generation**
System shows:
- “1 Suspicious File Detected”

Step 6 **: Report Generation**
User downloads:
- Evidence Report (CSV)
- Comparison Report

10. SYSTEM INTERFACE / IMPLEMENTATION

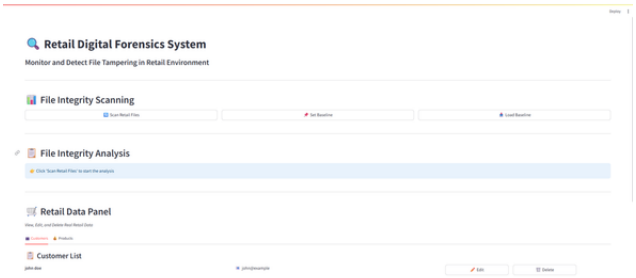

Figure 1: Main Dashboard of Retail Digital Forensics System

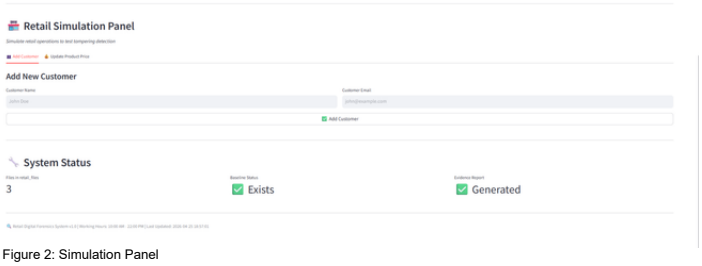

Figure 2: Simulation Panel

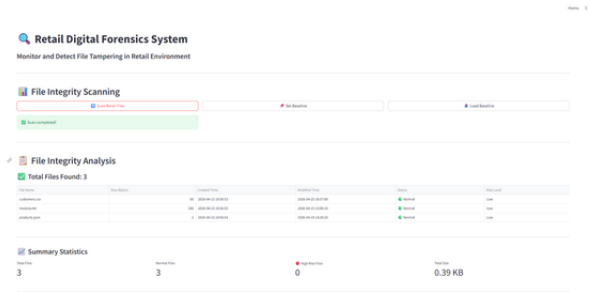

Figure 3: File Analysis Showing No Tampering Detected

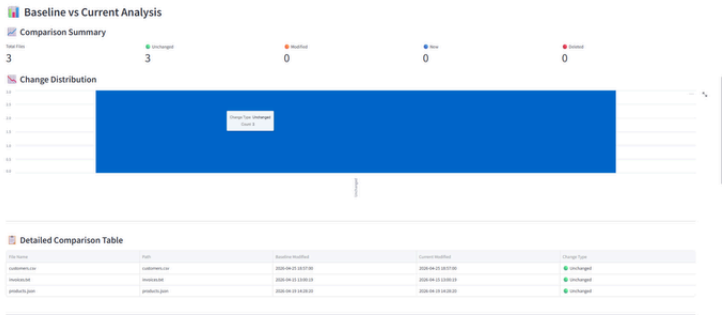

Figure 4: Baseline Comparison Showing No File Changes

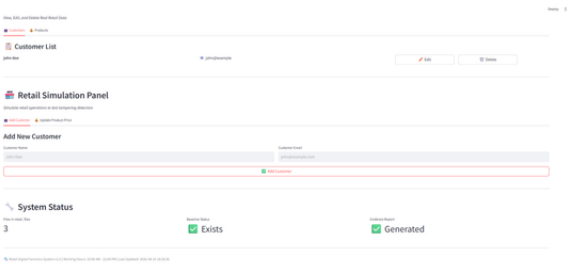

Figure 5: Simulation Panel Used to Modify Retail Data

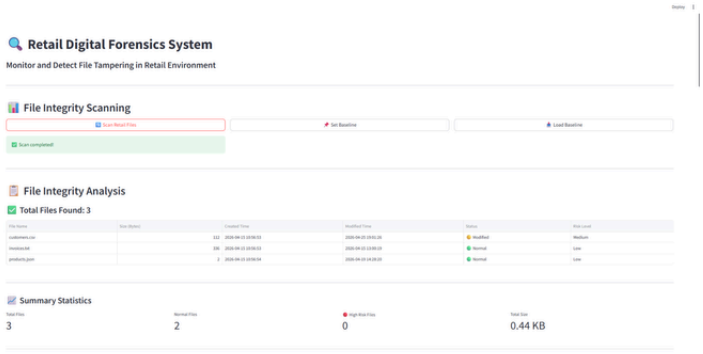

Figure 6: Detection of Modified File After Data Change

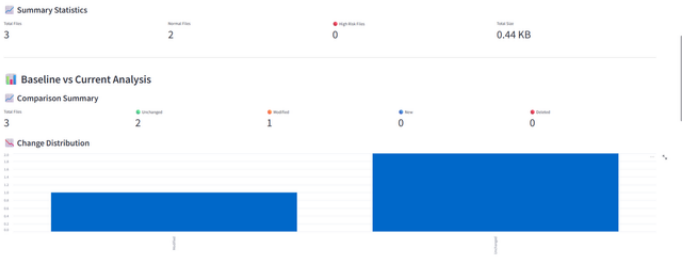

Figure 7: Baseline vs Current Showing Modified File

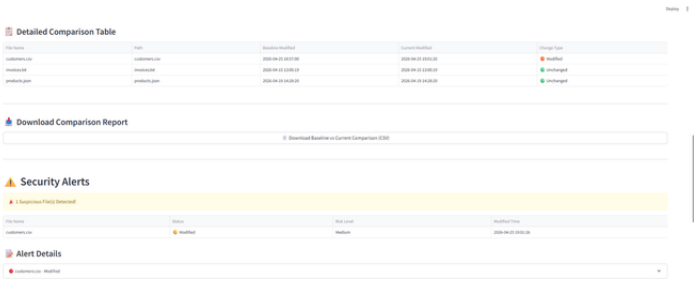

Figure 8: Security Alert for Detected Tampering

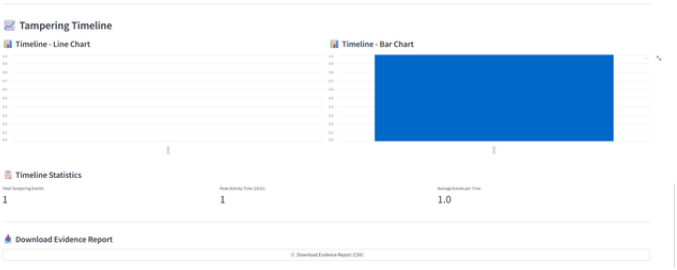

Figure 9: Timeline Visualization of Tampering Activity

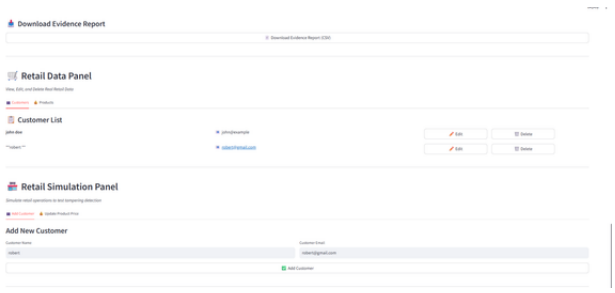

Figure 10: Retail Data Panel Showing Customer Management and Report Generation

	A	B	C	D	E	F	G	H
1	File Name	Path	Created Time	Modified Time	Status	Risk Level		
2	customers	C:\Users\J	15-04-2026 10:56	25-04-2026 19:01	Modified	Medium		
3	invoices.tx	C:\Users\J	15-04-2026 10:56	15-04-2026 13:00	Normal	Low		
4	products.j	C:\Users\J	15-04-2026 10:56	19-04-2026 14:28	Normal	Low		
5								
6								

Figure 11:Downloaded CSV file which shows the Created time and Modified time